

Walter Buga

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EDUCATION

- **Bachelor of Science, Informatics**
Arizona State University, School of Engineering

Work History

Software Engineer II at Arendai, inc.

August 2021 to current

- Created a warehouse simulation tool using Unity3D and C# to optimize and visualize warehouse operations.
- Used C#, JSON, and MongoDB to generate 3D facilities from warehouse data.
- Integrated with fleet orchestration systems via MQTT, VDA5050 protocol, Docker, and ROS2, enabling real-time control of 500+ autonomous robots across warehouse environments.
- Designed and developed large-scale 3D data visualization systems to monitor warehouse operations, transforming complex operational and robotics data into interactive real-time environments.
- Utilized MongoDB for efficient data storage and retrieval, supporting machine learning models for anomaly detection and predictive analytics.
- Integrated with backend services on AWS, ensuring high availability and scalability.
- Designed intuitive UI/UX features using XAML and UIToolkit to enhance user interaction and operational data visualization.
- Leveraged Git, GitHub, and CI/CD pipelines to automate testing, build deployment, and quality assurance.
- Implemented and integrated with REST APIs for data transfer between simulation components.
- Dockerized parts of Arendai's simulation stack for portability and scalability.
- Conducted PR reviews and mentored interns to ensure code quality, consistency, and team skill development.
- Showcased warehouse simulation technology at major U.S. tech conferences, including CES, leading to partnerships with major companies and increased industry adoption.

Technical Director at Dynamynd

November 2019 – September 2021

- Led a team of software developers to design and deploy high-quality, revenue-generating web applications with 20,000+ users, ensuring alignment with project goals and timelines.
- Maintained multiple web applications using JavaScript, HTML, & CSS. Optimized performance and ensured a seamless customer experience.
- Conducted technical interviews and resume evaluations, successfully hiring and onboarding top-tier developers, strengthening team capabilities and accelerating project execution.

Web Developer & Certified Client Specialist at Dynamynd

April 2017 – September 2019

- Upgraded web applications and websites using HTML, CSS, and JavaScript
- Developed Python scripts for data analysis that uncovered trends across 100K+ data points, guiding strategic business decisions.
- Built a python web scraper that sourced and parsed 10,000+ job descriptions that enabled the creation of a personalized career finder tool that improved user engagement and guidance.
- Handled 20+ high value clients by providing guidance in business, family, education, and other areas.
- Acted as CRM administrator, managing records for 20,000+ users, improving reporting accuracy and contact outreach.

Front-end Developer Intern at Dynamynd

December 2017 – March 2018

- Built and refined multiple websites in WordPress and ColdFusion, using HTML, CSS, and JavaScript.
- Collaborated with senior developers to apply best practices in maintainable, scalable UI development.

Front-end Developer Intern at Helix House

June 2017 – August 2017

- Created 5+ websites a week based on clients' needs using HTML, CSS, JavaScript and WordPress.
- Developed reusable front-end components with HTML/CSS to streamline UI creation across multiple sites.
- Partnered with creative directors to transform design specifications into pixel-perfect front ends.

PROJECTS

Offline AI Voice System

- Designed and implemented a fully offline AI NPC system in Unity & C#, integrating a local LLM (llama.cpp) and neural text-to-speech (Coqui TTS), enabling real-time conversational characters without cloud dependencies.
- Used Python to build an HTTP-based communication between Unity and local AI services, handling audio streaming, PCM16 WAV conversion, and low-latency playback via FMOD.
- Packaged and deployed the system as a portable build with auto-launching AI services, eliminating the need for external runtimes or user setup.

CHIP-8 3D Emulator

- Developed a CHIP-8 emulator in Unity using C#, implementing CPU instruction decoding, memory management, timers, stack operations, input handling, and real-time opcode execution.
- Implemented low-level systems concepts including bitwise operations, XOR sprite rendering, collision detection, and synchronized timing to successfully run classic CHIP-8 ROMs such as Pong and Tetris.

Procedural Warehouse Simulation

- Built a full-stack warehouse simulation system in Unity and C# featuring procedurally generated warehouse layouts and pathfinding for autonomous robot traversal.
- Implemented ASP.NET Core, PostgreSQL, SQL, REST APIs, and JSON serialization to save and load warehouse data.
- Designed a modular architecture separating simulation data, rendering, pathfinding, serialization, and backend communication; structured for future Docker Compose deployment with containerized ASP.NET and PostgreSQL services.

Augmented Reality App — Sponsored by Pizza Hut

- Developed an augmented reality iOS game for Pizza Hut using Unity3D, C#, and AR libraries.

Kanji Flash Card Web App

- Coded an interactive web application for learning Japanese Kanji, utilizing JavaScript, HTML, and CSS.

